

DEEP LEARNING — 046211
FINAL EXAM B
28/5/2022

EXAM DURATION: 3 HOURS

INSTRUCTIONS.

1. You can use a calculator and the 2 formula sheets (4 pages).
2. You need to answer all question parts, unless you received a special permit.
3. The weight of each question is written in the form. The total number of points is 100 points.
4. If you received a special permit (due to reserve duty) you can choose to answer only two questions out of three. In that case, the questions points will be normalized so the total number of points will remain 100.
5. Please write clearly and orderly. You can write either in English or in Hebrew.
6. You must give the full details of the solution derivation. Solutions without explanations will not be given a score, unless written otherwise.
7. Once you finish the exam, you need to use gradescope to upload your exam. In addition, you need to submit the notebook.

1. INVARIANCE AND EQUIVARIANCE IN MULTILAYER NEURAL NETWORKS

Shelly started working at the “Aperture” company, at the genetic testing department. She was asked to design an Multilayer Neural Network (MNN) with the following input: DNA sequences of length d . The DNA is a sequence of bases, where each base can be one of 4 options: (C, T, G, A) . Thus, the input can be described as the following matrix: $\mathbf{X} \in \mathbb{R}^{4 \times d}$, where $X[j, i]$ denotes the measured value of base concentration of the j -th base at location i .

The network should output a binary classification $y \in \{-1, 1\}$ for a specific property we wish to find. The network will be trained on samples $\{\mathbf{X}^{(n)}, y^{(n)}\}_{n=1}^N$, with a logistic loss function $\ell(y, \hat{y}) = \log(1 + e^{-y\hat{y}})$.

First, we will examine a network with one hidden layer of size $4 \times d$ and a LeakyReLU activation φ :

$$f_{\mathbf{w}}(\mathbf{X}) = \sum_{r=1}^4 \sum_{k=1}^d W_2[r, k] \varphi \left(\sum_{j=1}^4 \sum_{i=1}^d W_1[r, k, j, i] X[j, i] \right),$$

where $\mathbf{w} = \{\mathbf{W}_1, \mathbf{W}_2\}$ are the layers of the weight tensors. After training is done, the classification will be done with $\text{sign}(f(\mathbf{X}))$.

1. Which invariances exist in the network’s parameters?
2. Now, Shelly noticed that the direction in which the DNA is scanned is arbitrary. Thus, if for two inputs $\mathbf{X}, \tilde{\mathbf{X}}$ we have $\forall i, j : \mathbf{X}[j, i] = \tilde{\mathbf{X}}[j, d - i + 1]$, then the two inputs are equivalent in their meaning. What constraints should we put on the network’s parameters to improve the network’s classification performance? Explain why using an invariant hidden layer is not optimal.
3. Lastly, Shelly noticed that the measurement process is noisy, so each sample $\mathbf{X}^{(n)}$ is multiplied by an arbitrary scale. Thus if for two inputs $\mathbf{X}, \tilde{\mathbf{X}}$ we have $\forall i, j : \mathbf{X}[j, i] = c\tilde{\mathbf{X}}[j, i]$, for some constant $c > 0$, then the two inputs are equivalent in their meaning.
 - (a) For the given network, that is already trained, what is the effect of the scale c on the classification result?
 - (b) Can the arbitrary scale hurt the training process? Hint: think what happens to the gradient of each sample.
 - (c) How can we use this information to improve the classifier performance?

2. EFFICIENT DIFFERENTIATION

We wish to optimize a loss function $\mathcal{L}(\mathbf{w})$ for $\mathbf{w} \in \mathbb{R}^d$ using Gradient Descent (GD) with some step size schedule η_t

$$(2.1) \quad \forall t = 1, 2, \dots : \mathbf{w}(t) = \mathbf{w}(t-1) - \eta_t \nabla \mathcal{L}(\mathbf{w}(t-1))$$

initialized from some $\mathbf{w}(0)$. We would like to learn the best step size schedule using GD.

1. Suppose we can consider each η_t as a separate parameter for each t . We initialize this parameter with η_0 and update η_{t-1} with a GD step on $\mathcal{L}(\mathbf{w}(t-1))$

$$(2.2) \quad \eta_t = \eta_{t-1} - \alpha_t \frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-1}}$$

for every step of eq. 2.1, where α_t is the another step size.

Calculate $\partial \mathcal{L}(\mathbf{w}(t-1)) / \partial \eta_{t-1}$ as a function of the loss gradients $\nabla \mathcal{L}(\mathbf{w}(t-1))$ and $\nabla \mathcal{L}(\mathbf{w}(t-2))$.

2. Now we wish to update (η_{t-1}, η_{t-2}) by doing a GD step on $\mathcal{L}(\mathbf{w}(t-1))$

$$(2.3) \quad (\eta_{t+1}, \eta_t) = (\eta_{t-1}, \eta_{t-2}) - \alpha_t \left(\frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-1}}, \frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-2}} \right)$$

every two steps of eq. 2.1.

Calculate the derivative $\frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-2}}$ as a function of $\eta_{t-1}, \{\nabla \mathcal{L}(\mathbf{w}(t-k))\}_{k=1}^3$ and $\nabla^2 \mathcal{L}(\mathbf{w}(t-2))$.

3. Now we wish again to update $(\eta_t, \eta_{t+1}, \dots, \eta_{t+T})$ by doing a GD step on $\mathcal{L}(\mathbf{w}(t-1))$ every T steps of eq. 2.1

$$(2.4) \quad (\eta_{t+T}, \dots, \eta_t) = (\eta_{t-1}, \dots, \eta_{t-1-T}) - \alpha_t \left(\frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-1}}, \dots, \frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-1-T}} \right)$$

Calculate the derivative $\frac{\partial \mathcal{L}(\mathbf{w}(t-1))}{\partial \eta_{t-\tau}}$ as a function of $\{\eta_{t-k}, \nabla^2 \mathcal{L}(\mathbf{w}(t-k-1))\}_{k=1}^{\tau-1}$, $\nabla \mathcal{L}(\mathbf{w}(t-1))$ and $\nabla \mathcal{L}(\mathbf{w}(t-\tau-1))$.

4. Compare this approach (eq. 2.4 with $T > 1$) to the first one (eq. 2.2). Name one advantage for each approach. Hints: Think of computational complexity, ease of optimization, suitability of the objective.

3. “TYPICAL” GENERALIZATION IN MULTILAYER NEURAL NETWORKS

We examine a “student” neural network $f_{\mathbf{w}}(\mathbf{x})$ with parameter vector $\mathbf{w} \in \mathbb{R}^k$ and input $\mathbf{x} \in \mathbb{R}^{d_0}$ used in a binary classification problem where the training set is $\mathcal{S} = \{\mathbf{x}^{(n)}\}_{n=1}^N$ sampled i.i.d. from P_X , where the binary (± 1) labels are generated by a “teacher” neural network $f_{\mathbf{w}_*}(\mathbf{x})$ with the same architecture. To understand the “typical” generalization of the student, we examine the following “Guess and Check” algorithm to learn its weights: we randomly sample parameters vectors $\mathbf{w}_1, \mathbf{w}_2, \dots$ i.i.d. from P_W , in which each parameter is sampled independently from a uniform distribution over $Q = \{-(q-1)/2, \dots, -1, 0, 1, \dots, (q-1)/2\}$ quantization levels, where $q = |Q|$ is an odd positive number. We do this until a stopping time t in which we perfectly fit the dataset: $\forall n : f_{\mathbf{w}_t}(\mathbf{x}^{(n)}) = f_{\mathbf{w}_*}(\mathbf{x}^{(n)})$. We examine a two-layer neural network with d_1 hidden neurons

$$f_{\mathbf{w}}(\mathbf{x}) = \text{sign}(\mathbf{w}_2^\top [\mathbf{W}_1 \mathbf{x}]_+)$$

where $[\cdot]_+$ is the ReLU activation function, the teacher has at most $d_1^* < d_1$ non-zero neurons (i.e., the other $d_1 - d_1^*$ hidden neurons in the teacher to have all the incoming and outgoing weights equal to zero).

1. Calculate the probability $P_{\mathbf{w} \sim P_W}(\mathbf{w} = \mathbf{w}_*)$.

2. Prove that

$$(3.1) \quad p_* \triangleq P_{\mathbf{w} \sim P_W}(\forall \mathbf{x} : f_{\mathbf{w}}(\mathbf{x}) = f_{\mathbf{w}_*}(\mathbf{x})) \geq q^{-d_0 d_1^* - d_1}.$$

3. Show that for any constant $T > 0$, we can bound the probability of stopping time $t > T$ as

$$(3.2) \quad [T] \leq \frac{\log P(t > T)}{\log(1 - p_*)}.$$

4. Prove the generalization bound

Theorem 1. *With probability $(1 - \eta)(1 - \delta)$,*

$$(3.3) \quad \epsilon < \frac{(d_0 d_1^* + d_1) \log q + \log \frac{1}{\delta} + \log \log \frac{1}{\eta}}{N}$$

Hint: Combine the results from previous sections, using the approximations $[T] \approx T$ and $\log(1 - p_*) \approx -p_*$ (treat these approximations as exact), and the following basic generalization bound (which we learned in class)

Theorem 2. *For any $f \in |\mathcal{F}|$, with probability $1 - \delta$,*

$$(3.4) \quad \epsilon \triangleq \mathbb{P}_{\mathbf{x}}(f_{\mathbf{w}}(\mathbf{x}) \neq f_{\mathbf{w}_*}(\mathbf{x})) < \frac{\log |\mathcal{F}| + \log \frac{1}{\delta}}{N}.$$

5. Is the bound in eq. 3.3 better than the bound in eq. 3.4 in which $\mathcal{F} = \{f_{\mathbf{w}} : \mathbf{w} \in Q^k\}$ is the student hypothesis class (in which each parameter can have one of q values)? Explain and ignore the (negligible) $\log \log \frac{1}{\eta}$ term.